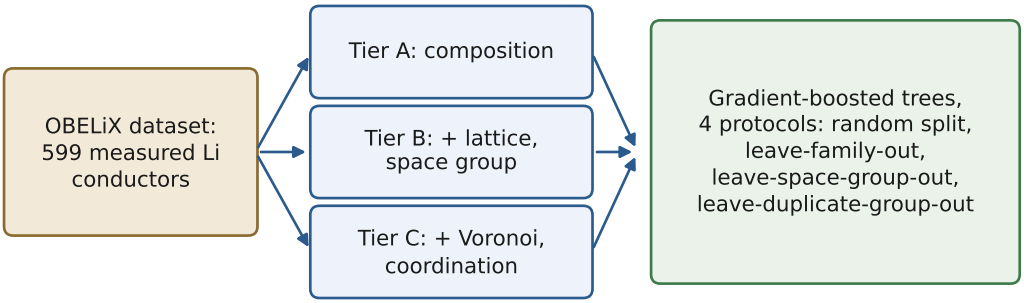
Composition versus structure in ML models of ionic conductivity
OBELiX dataset: 599 measured Li conductors
Tier A: composition
Tier B: + lattice, space group
Tier C: + Voronoi, coordination
Gradient-boosted trees, 4 protocols: random split, leave-family-out, leave-space-group-out, leave-duplicate-group-out
Key findings
Composition alone: MAE 1.86, R^2 0.29; below published baseline (1.59)
+ lattice, space group: MAE 1.62, matches baseline; Voronoi adds nothing on 3 of 4 protocols
Screening: top-30% enrichment about 1.9; top-10/20% equals chance